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IoT based system for greenhouses remote monitoring and climate control

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***Abstract*—The proposed system incorporates a network of sensors strategically placed within the greenhouse to collect crucial environmental data, including temperature, humidity and soil moisture. These sensors form a connected ecosystem that continuously transmits data to a cloud-based platform. The cloud infrastructure, implemented using IBM Watson IoT platform, serves as a centralized hub for data storage, analysis, and decision-making. Furthermore, the system includes actuators responsible for controlling environmental parameters within the greenhouse. Automated climate control mechanisms are orchestrated based on real-time data insights. The implementation utilizes Raspberry Pi as the central processing unit, running Python scripts to interact with the sensors and actuators. To facilitate user-friendly interaction and real-time monitoring, the system integrates web interfaces developed using Node-Red. These interfaces provide remote access to greenhouse conditions and enable users to make informed decisions for optimal plant growth. An automated greenhouse prototype is finally implemented for monitoring and climate control system.**

Keywords—Smart greenhouse, Internet of Things, sensors, actuators, Raspberry board, cloud platform, web application.

I. INTRODUCTION

The Internet of Things (IoT) emerges as a transformative force with the potential to address multiple challenges [1-5]. The agricultural landscape is undergoing a transformative shift fueled by advancements in technology, and the Internet of Things (IoT) stands at the forefront of this revolution [6-10]. Greenhouse cultivation has long been a cornerstone of modern agriculture, allowing for controlled and optimized growing environments. However, the traditional methods of greenhouse management are increasingly being augmented by IoT technologies to meet the demands of precision farming. In this sense, the integration of IoT in greenhouse systems holds the promise of enhancing productivity, minimizing resource wastage, and providing real-time insights for informed decision-making [11-20].

In response to the growing challenges faced by the agricultural sector, such as the need for resource optimization, sustainable practices, and remote monitoring, this paper introduces an innovative IoT-based system for greenhouses remote monitoring and climate control. Indeed, we delve into the design, implementation, and functionality of a sophisticated IoT-based system tailored specifically for remote monitoring and climate control in greenhouses. By leveraging a network of sensors strategically deployed within the greenhouse environment, the system aims to capture vital data points such as temperature, humidity and soil moisture. These data are then seamlessly transmitted to a cloud-based platform, enabling centralized processing, storage, and analysis. The backbone of our proposed system lies in its ability to translate real-time data into actionable insights. Automated actuators, including mechanisms for heating, ventilation and irrigation are orchestrated based on the continuous stream of data. The central processing unit, in the form of a Raspberry Pi, executes Python scripts to interact with the sensors and actuators, creating a responsive and dynamic ecosystem within the greenhouse. As precision agriculture becomes increasingly integral to modern farming practices, our IoT-based system addresses the critical need for efficient and remote greenhouse management. The implementation also embraces user-friendly interfaces developed using Node-Red, offering a comprehensive platform for real-time monitoring and decision-making from any location. Accordingly, this paper aims to explore the intricate interplay between IoT technology and greenhouse management, shedding light on the transformative potential it holds for sustainable and optimized agriculture practices. Accordingly, through a detailed examination of the system's architecture, functionalities, and real-world application, we contribute to the evolving narrative of smart agriculture and its pivotal role in shaping the future of food production.

This research is structured into four distinctive sections. The second section presents the concept for IoT powered greenhouse monitoring and climate control system. Moving forward, the third section delves into the creation of a fully automated greenhouse prototype, showcasing the practical implementation of the monitoring and climate control system. The final section conclude the study, encapsulating and summarizing the key discoveries throughout the research findings.

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II. Architecture

Our project consists of a connected greenhouse, within which a greenhouse control module and plant-level sensors modules are installed. It also includes a supervision and control module. Access to the cloud for various modules enables a safe and secure exchange of data.

The architecture of the designed smart greenhouse is described in the following Fig. 1.

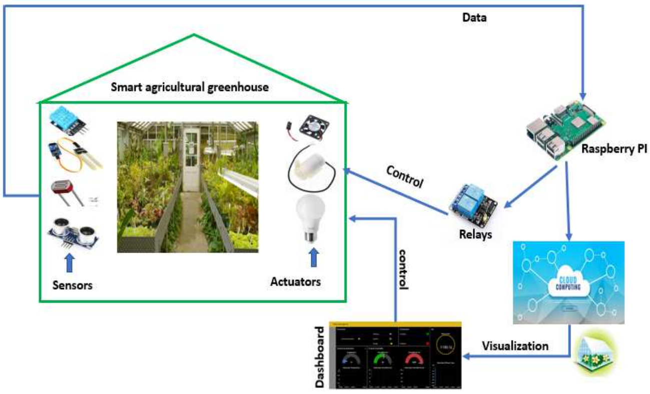

Fig. 1. The design of the proposed IoT system

The sensors modules are designed to perform diverse measurements, including temperature, humidity and soil moisture. These modules are capable of remote communication to send measurement data to the cloud. The control module is tasked with overseeing the diverse actuators within the greenhouse. It needs to possess remote communication capabilities to gather pertinent measurement information crucial for the automated control process.

A. *Hardware devices*

The hardware components of the designed smart agricultural greenhouse form a comprehensive system designed to enhance efficiency and productivity. At its core is the Raspberry Pi, a compact yet powerful computer serving as the central processing unit. Complementing this, the DHT11 sensor monitors humidity and temperature in real-time, offering crucial insights into the greenhouse environment. The soil moisture sensor ensures precise irrigation by tracking soil moisture levels. Additionally, the LDR sensor gauges light changes, contributing to the optimization of the overall greenhouse conditions. The SRF05 Sensor, functioning ultrasonically, facilitates diverse applications such as monitoring plant growth and detecting water levels in reservoirs. Acting upon this data are various actuators, including the water pump for controlled irrigation, the fan regulating airflow and temperature, and relays serving as switches for device control. Together, these hardware components synergize to establish a smart agricultural greenhouse system adept at monitoring and fine-tuning environmental parameters to foster optimal plant growth. The Raspberry Pi acts as the system's command center, processing sensors data and orchestrating actuators functions to maintain the desired greenhouse conditions.

1) *Raspberry board*

The Raspberry board functions as a microcontroller, adeptly handling data from sensors integrated into the intelligent greenhouse. It plays a pivotal role in gathering, organizing, and wirelessly transmitting data related to gait.

2) *DHT11 sensor*

The DHT11 sensor provides real-time data about humidity and temperature. This information is crucial for plants' well-being as it enables the smart system to adjust parameters like ventilation, irrigation, and temperature control.

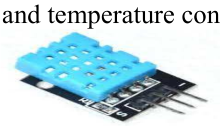

Fig. 2. The DHT11 sensor

3) *Soil moisture sensor*

This sensor plays a crucial role in optimizing irrigation practices and ensuring that plants receive an appropriate amount of water.

Fig. 3. The soil moisture sensor

4) *LDR sensor*

The Light Dependent Resistor (LDR) sensor is sensitive to light, and its resistance changes based on the intensity of light falling on it. In the designed smart greenhouse, the LDR sensor is employed to monitor and control aspects related to light conditions.

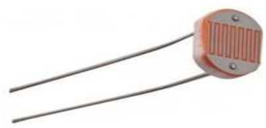

Fig. 4. The LDR sensor

5) *SRF05 sensor*

By providing accurate data on distances, the SRF05 sensor assists in tasks such as monitoring plant growth and detecting water levels in reservoirs.

Fig. 5. The SRF05 sensor

6) *Water pump*

A water pump, designed to move water, is essential for this purpose. In our setup, we employ a compact water pump constructed with a noise-free brushless motor for silent and efficient operation.

Fig. 6. The water pump

7) *Fan*

A fan is used to offer the flexibility to introduce or expel air efficiently. Moreover, if we opt for bringing in fresh air with open windows, it would naturally renew the air inside the greenhouse. Our preference aligns with a straightforward fan solution, as depicted in the accompanying figure.

Fig. 7. The fan

B. *Software devices*

The software components of the designed smart agricultural greenhouse include essential tools for programming, data analysis, cloud integration, and remote access. The Python Integrated Development Environment (IDE) serves as a programming platform, facilitating the development and execution of Python scripts for various automation tasks within the greenhouse system. IBM Watson, a robust artificial intelligence and analytics platform, contributes advanced data analysis capabilities, enabling the greenhouse to make informed decisions based on collected data. Furthermore, IBM Cloud provides a cloud computing environment, allowing for the storage, processing, and retrieval of data from the smart greenhouse. Node-Red, a flow-based development tool, enhances the automation process by providing a visual programming interface for connecting devices and services within the greenhouse system. SSH (Secure Shell) and VNC (Virtual Network Computing) are included for remote access, enabling users to connect to and control the smart agricultural greenhouse from a distance, enhancing accessibility and management capabilities.

III. RESULTS

This section is structured into four distinct subsections: the initial one embodies the greenhouse and its internal devices, the second delves into the realm of software devices, the third subsection focuses on the realm of real-time data, and the ultimate subsection is dedicated to the intricacies of the crafted prototyping.

A. *Hardware configuration*

The hardware configuration of the designed smart agricultural greenhouse is intricately designed, encompassing a range of physical components that collectively establish the foundation of the whole intelligent system. At its core lies the Raspberry Pi microcontroller, functioning as the central processing unit and serving as the brain of the greenhouse. This pivotal component manages the diverse devices and processes critical for the system's operation. Key sensors, including the DHT11 for real-time monitoring of temperature and humidity, the soil moisture sensor for efficient irrigation, the LDR (Light Dependent Resistor) sensor for measuring light changes, and the SRF05 sensor for ultrasonic distance detection, provide crucial environmental data. Complementary actuators, such as the water pump for irrigation control, the fan for regulating airflow and temperature, and relays acting as switches based on smart system decisions, contribute to the system's responsive functionality. Additionally, the incorporation of a well-designed ventilation system ensures optimal air circulation—a fundamental factor for maintaining the health of the greenhouse plants.

The complete diagram of the designed system is summarized in the following Fig. 8.

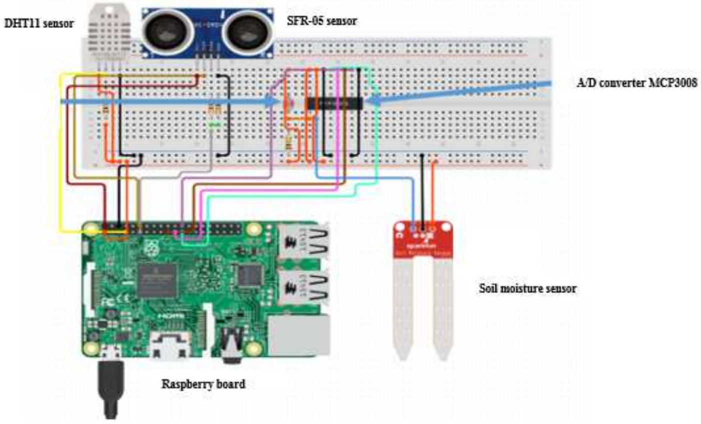

Fig. 8. The designed smart insole

The designed smart agricultural greenhouse based on IoT devices is presented in the following Fig. 9.

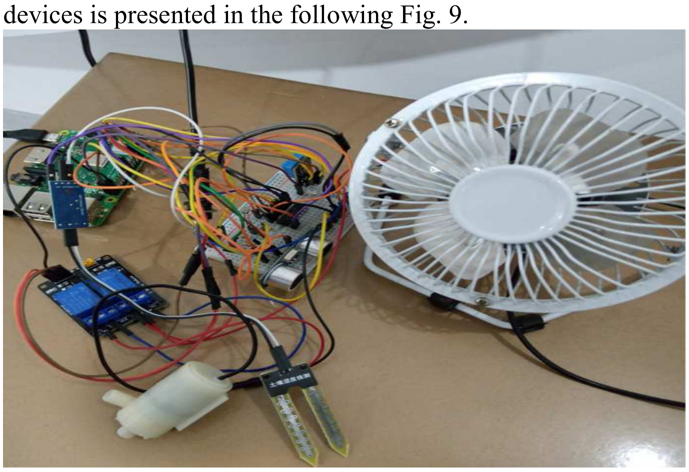

Fig. 9. The designed smart insole

B. *Software configuration*

The software components of the designed smart greenhouse integrate a variety of programming languages and platforms to establish a robust system for monitoring, control, and automation. Python, recognized for its versatility and potency, is employed for scripting and automating tasks within the smart greenhouse. Its readability and extensive libraries make it an ideal choice for interfacing with diverse hardware components. The utilization of IBM Cloud harnesses the capabilities of cloud computing, providing a platform for efficient data storage, processing, and analysis. This integration facilitates seamless connections with other services and enables remote access to greenhouse data. Node-RED, functioning as a visual programming tool, simplifies the creation of flow-based application, allowing users to establish connections between devices, sensors, and services, thereby streamlining the automation process. The inclusion of SSH (Secure SHell) ensures secure remote access to the greenhouse system, providing a protected communication channel for managing and monitoring the system from a distance. Additionally, VNC (Virtual Network Computing) serves as a remote desktop access tool, empowering users to visualize and control the graphical user interface of the smart greenhouse system remotely. Together, these software components work in harmony to guarantee efficient communication, data processing, and user interaction within the smart greenhouse, forming a comprehensive and effective software ecosystem for its optimal operation.

1) VNC configuration

To initiate this process, the first step involves connecting to the Raspberry Pi through SSH (Secure SHell), a secure communication protocol and software. SSH ensures the encryption of all executed commands, maintaining the confidentiality of operations. Establishing an SSH connection is highly practical particularly for programming with Node-RED. Indeed, SSH allows remote connection to the machine using a user account. To facilitate this, we should provide a username and a password. We are then required to insert the MicroSD card of the Raspberry Pi into the computer, navigate to the card, and create a file named "ssh" in the boot partition. Once SSH is established, the VNC server, pre-installed on Raspbian, can be activated.

Fig. 10. SSH activation

To accomplish this, simply run the command **sudo raspi-config**, navigate to Interfacing Options, and then choose VNC from the options.

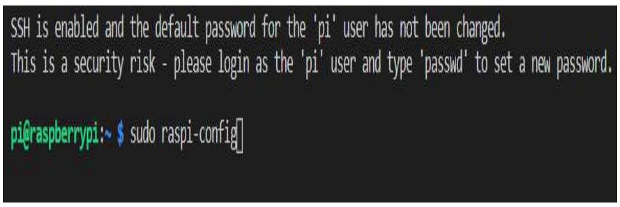

Fig. 11. VNC activation

The VNC server is then activated on the Raspberry Pi, the next step is to connect to it. Firstely, a client on the computer is installed. By default, the Raspberry Pi uses the RealVNC server, configured to employ a user account connection mode. Once the RealVNC viewer is installed, we have to launch it and to connect to the Raspberry Pi IP address.

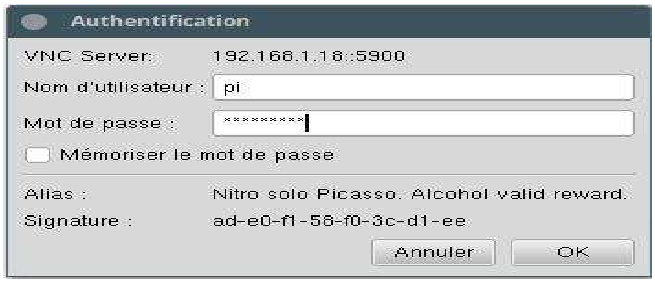

Fig. 12. Connexion via RealVNC

The connection to an IP address is now established and then it is recorded by RealVNC Viewer. We can then configure various options such as image quality, fluidity, and more.

2) IBM Cloud configuration

A free account on IBM Cloud is created with a lite plan that provides free access to 500 devices, 500 application links, and 200 MB per month of exchanged data. The IBM Watson IoT Platform is then created. Afterward, two categories of devices are introduced, one for the sensors modules and the other for the control module. For each type of device, we configure it with MQTT (Message Queue Telemetry Transport) access, ensuring it could authenticate with the organization ID, device type, device ID, and authentication token.

MQTT is an extremely simple and lightweight messaging protocol for data exchange.

3) Design of the supervision module

The supervision module consists of two parts:

Monitoring the greenhouse's status

It corresponds to the development of web interfaces which are driven by the Node-Red web application

Below represents the various nodes used in our platform to display plant data.

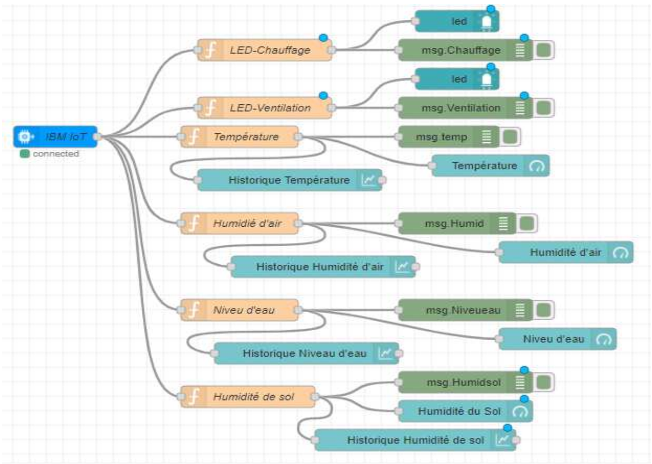

Fig. 13. Node-Red diagram of the greenhouse monitoring

In this diagram, we can discern:

- The "IBM IoT" node which is an input node used with the Watson IoT Platform to receive events sent by devices, receive commands sent to devices, or receive updates on the state of devices or applications.
- Function nodes (temperature, humidity, soil humidity, water level…) that read data from the cloud sent by sensors.
- Gauge and chart nodes are used to graphically display these values on a web interface.

Remote control

The diagram showcased in the following Fig. 14 delineates the remote configuration of automatic control parameters within the control module and the remote operation of the greenhouse's actuators.

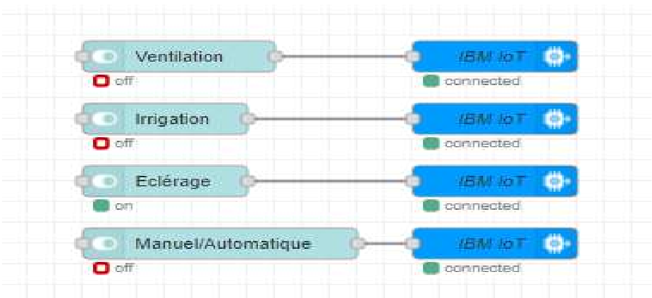

Fig. 14. Node-Red Diagram of the remote control

This diagram highlights:

- The "IBM IoT" node, an output node integrated with the Watson IoT Platform for issuing commands or sending events.
- The "switch" nodes, which present a switch on the user interface. Each alteration in the switch's state generates a message on the output with the specified activation/deactivation value.

C. *Prototyping*

To evaluate the system's real-life performance, we successfully constructed a prototype as shown in the following Fig. 15. In order to automate and oversee our mini greenhouse, the initial step involve selecting the dimensions of the work surface. It is essential to securely position sensors and actuators within the greenhouse while facilitating the passage of connection cables. Moreover, the greenhouse needed to provide ample space for at least two plant pots and support a plant height of up to 30cm. We opte for a 40*60 cm wooden frame with collapsible water pipes, covered by a green mosquito net.

Fig. 15. The prototype of the designed greenhouse

1) *Web interface and data measurement*

The following Fig. 16 illustrates the enhanced web interface created for monitoring greenhouse conditions.

Fig. 16. Web interface generated for greenhouse monitoring

The generated web interface is specifically designed to facilitate comprehensive monitoring of various parameters within the greenhouse environment. This user-friendly interface provides real-time data on factors such as temperature, humidity, and other relevant metrics (water level…). The development of this web interface is a crucial component of the greenhouse monitoring system, offering users a seamless and intuitive platform to efficiently manage and analyze the data collected from differents sensors.

Additional diverse measures implemented are depicted in the accompanying Fig. 17.

Fig. 17. Different measures taken

2) *Sending a notification through Email*

This entails gathering data from the greenhouse through sensors and sending it to a personal Gmail account via Email. Prior to proceeding, it is essential to generate an "application password" using a Google account, which is then inserted into IBM to enable secure access for the connection.

As shown in the following Fig. 18, a function in Node-RED is established with a condition linked to the water level in the tank. If it drops below 0.5, an email alert is triggered.

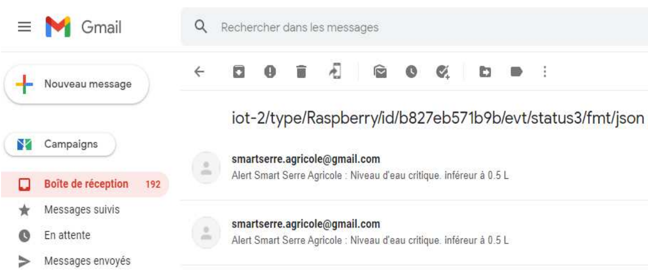

Fig. 18. Remote control by using Email

IV. CONCLUSION

The agricultural sector remains a fundamental priority for humanity, and contemporary advancements in technology, especially the Internet of Things (IoT), offer the potential to achieve food production that can adequately cater to the needs of global populations. The IoT, having already proven its efficacy across various sectors, holds considerable promise for improving food production both quantitatively and qualitatively. In this context, a prototype for an intelligent greenhouse using a cloud-based IoT approach is designed in this research work. The system design focuses on three microclimatic parameters within the greenhouse: humidity level, temperature, and water level. These parameters undergo regular monitoring and maintenance through the deployment of sensors modules at the plant level and a control module (responsible for heating, ventilation, irrigation, and lighting) within the greenhouse. These modules are crafted around the embedded Rosebery PI platform and the Python programming language. They are consistently linked to the IBM Watson cloud through its IoT platform. The capability for remote monitoring and control of the greenhouse is facilitated by web interfaces integrated with the cloud. These interfaces are developed and generated using the Node-Red environment.

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